6/3/2026

Finding PI

1. RHS, $X = e^{ax}$

P.I is:

$$y_p = \frac{1}{f(D)} e^{ax}$$

$$= \begin{cases} \frac{1}{f(a)} e^{ax}, & \text{if } f(a) \neq 0 \\ \frac{x}{f'(a)} e^{ax}, & \text{if } f(a) = 0,\ f'(a) \neq 0 \\ \frac{x^2}{f''(a)} e^{ax}, & \text{if } f(a) = 0,\ f'(a) = 0,\ f''(a) \neq 0 \\ \vdots & \\ \frac{x^n}{f^{(n)}(a)} e^{ax}, & \text{if } f(a) = 0,\ f'(a) = 0, \dots f^{n-1}(a) = 0,\ f^{(n)}(a) \neq 0 \end{cases}$$

$$p_0 \frac{d^n y}{dx^n} + p_1 \frac{d^{n-1} y}{dx^{n-1}} + \dots + p_{n-1} \frac{dy}{dx} + p_n y = X$$

$$D = \frac{d}{dx}$$

$$\Rightarrow (p_0 D^n + p_1 D^{n-1} + \dots + p_{n-1} D + p_n) y = X$$

3. Solve $6 \frac{d^2 y}{dx^2} + 17 \frac{dy}{dx} + 12 y = e^{-\frac{3x}{2}} + 2^x$

$e^{\log 2^x}$ ⚠!

Soln: The auxiliary eqn is:

$$6D^2 + 17D + 12 = 0, \text{ where, } D = \frac{d}{dx}$$

$$\Rightarrow D = -\frac{4}{3}, -\frac{3}{2}$$

$\therefore$ C.F is $y_c = c_1 e^{-\frac{4}{3}x} + c_2 e^{-\frac{3}{2}x}$

$$(p_0 D^n + p_1 D^{n-1} + \dots + p_{n-1} D + p_n) y_c = 0 \quad \text{--- (1)}$$

$$\& \ (p_0 D^n + p_1 D^{n-1} + \dots + p_{n-1} D + p_n) y_p = X \quad \text{--- (2)}$$

(1) + (2)

$$\Rightarrow (p_0 D^n + \dots + p_{n-1} D + p_n)(\underbrace{y_c + y_p}_{y}) = X$$

Now to get PI:

$$RHS = e^{-\frac{3x}{2}} + 2^x$$

$$= e^{-\frac{3x}{2}} + e^{\log 2^x}$$

$$\Rightarrow RHS = e^{-\frac{3x}{2}} + e^{x(\log 2)}$$

Now,

$$y_c = \frac{1}{f(D)}\left[e^{-\frac{3x}{2}} + e^{x(\log 2)}\right]$$

Here $f(D) = 6D^2 + 17D + 12$

[Always we need to solve

$f(D) \cdot y = X \Rightarrow y = \frac{1}{f(D)} \cdot X$

Aux. eqn: $f(D) = 0$]

$$\Rightarrow y_c = \frac{1}{6D^2 + 17D + 12}\left[e^{-\frac{3x}{2}} + e^{x(\log 2)}\right]$$

Consider: $e^{-\frac{3x}{2}}$

Here $f(D) = 6D^2 + 17D + 12$

$a = -\frac{3}{2}$, $f(-\frac{3}{2}) = 0$

Now $f'(D) = 12D + 17$

$f'(a) = f'(-\frac{3}{2}) = -1$

$$\therefore y_p \text{ (first term)} = \frac{x}{f'(-\frac{3}{2})} e^{-3/2\, x}$$

$$= -x \cdot e^{-3/2\, x}$$

consider: $e^{x(\log 2)}$

Here $f(D) = 6D^2 + 17D + 12$

$a = \log 2$

& $f(\log 2) = 6(\log 2)^2 + 17(\log 2) + 12$

$\neq 0$ $[= 26.66..]$

$$\therefore y_p \text{ (2nd term)} = \frac{1}{6(\log 2)^2 + 17(\log 2) + 12} \cdot e^{x(\log 2)}$$

$$\therefore y_p = -x \cdot e^{-3/2\, x} + \frac{1}{6(\log 2)^2 + 17(\log 2) + 12} \cdot e^{x(\log 2)}$$

The general soln is: $y = y_c + y_p$

Case 2 : When RHS function X is:

$\sin(ax+b)$ or $\cos(ax+b)$

Procedure for finding P.I.

Let the given d.e be

$$f(D)y = \sin(ax+b) \,/\, \cos(ax+b)$$

$$\Rightarrow y_p = \frac{1}{f(D)}\left[\sin(ax+b) \,/\, \cos(ax+b)\right]$$

$$= \begin{cases} \dfrac{1}{f(-a^2)} \sin(ax+b) \,/\, \cos(ax+b) & \text{if } f(-a^2) \neq 0 \\ \dfrac{x}{f'(-a^2)} \sin(ax+b) \,/\, \cos(ax+b) & \text{if } f(-a^2)=0,\ f'(-a^2) \neq 0 \\ \vdots & \vdots \\ \dfrac{x^n}{f^{(n)}(-a^2)} \sin(ax+b) \,/\, \cos(ax+b) & \text{if } f(-a^2)=0,\ f'(-a^2)=0, \ldots f^{(n-1)}(-a^2)=0,\ f^{(n)}(-a^2) \neq 0 \end{cases}$$

Here we replace D^2 by $-a^2$

eg: $X = \sin(3x+2)$

$f(D) = D^2 + 7D + 3$

$a = 3$

$f(-3^2) = -3^2 + 7D + 3$

$= -9 + 7D + 3 = 7D - 6$

$f(D) = 3D^4 + 2D^3 + D^2 + 12D$

$(D^2)^2$

$a = 3$

$f(-a^2) = 3(-3^2)^2 + 2D(-3^2) + (-3^2) + 12D$

Egs

1. Find the P.I of $(D^4 + D^2 + 1)y = \sin 2x$

Solⁿ: P.I is:

$$y_p = \frac{1}{f(D)} \sin 2x$$

Where $f(D) = D^4 + D^2 + 1 = (D^2)^2 + D^2 + 1$

$a = 2$

$f(-(2)^2) = (-2^2)^2 + (-2^2) + 1 = 16 - 4 + 1$
$= 13 \neq 0$

[Replacing D^2 by -2^2]

$$\boxed{-(2)^2 \neq (-2)^2}$$

$$\therefore \boxed{y_p = \frac{1}{13} \sin 2x}$$

2. Solve $(D^2 - D + 1)y = \cos 2x$

Solⁿ: